Ensemble-based Probabilistic Subsurface Facies Propagation Using Machine Learning and Multi-Attribute Analysis

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Introduction

Seismic facies analysis is a fundamental component of subsurface reservoir characterization in petroleum geoscience. Traditional facies mapping relies heavily on sparse well data, limiting spatial coverage and introducing significant interpolation uncertainties between wellbores. The integration of 3D seismic data with machine learning offers a innovative approach to propagate facies information across entire survey volume.

Seismic attributes which are mathematical transformations of seismic trace amplitudes, capture distinctive geological and petrophysical characteristics that correlate with rock properties, fluid content, and depositional environments. When combined with supervised learning algorithms trained on well-constrained facies distribution, these attributes enable predictive modeling that extends beyond direct well control.

Problem Statement

The primary challenge addressed in this project is the propagation of facies classifications from limited well log data to the full 3D seismic volume. Specific technical challenges include:

- Accurate prediction and spatial propagation of subsurface facies from seismic data is challenging.
- Well logs provide facies at discrete points, but lateral propagation in seismic volume is uncertain.
- Need for a data driven, automated approach to integrate seismic attributes and well facies.

Objective

To create a 3D facies model using Multi-Attribute seismic and well log data.

Methodology

Data Used: This project uses two main types of the data: Seismic and well log data, both essential for facies analysis and attribute based interpretation.

Seismic Data:

- Format: SEG-Y (Society of Exploration Geophysicists Y format)

- Primary Volume**: ST8511792.segy** representing time domain 3D seismic
- Domain: Two-way travel time (TWT)
- Velocity Data: **Average_Velocity_cube.segy** used for time-to-depth conversion

Well Data:

- Multiple well logs were used in this project, providing detailed information on facies and depth variation.
- Wells logs are stored in the **Well-Data-Analysis/Wells/** folder
- Data is used for attribute extraction, model validation and facies-classification.

Data Preprocessing Steps

TRACE-HEADER-INFO-time-domain.ipynb

- Read and checked **time-domain seismic headers**
- Verified **inline, crossline, coordinates, and time sampling** before depth conversion.

TRACE-header-depth-domain.ipynb

- Updated headers after **depth conversion**.
- Corrected **depth sampling, coordinates, and elevation**.

1. Well Data Analysis

We used the notebook **well-plot.ipynb** to visualize well logs such as gamma ray, porosity, permeability, facies, and net-to-gross. The Wells folder contained all the well log files in LAS formats.

2. **Depth-domain conversion**:

To properly align the seismic data with well logs, we converted the seismic cube from the **time domain** to the **depth domain**.

For this, we used the **Average Velocity Cube (Average_Velocity_cube.segy)** to apply time-to-depth conversion.The final output was a depth-aligned 3D seismic cube named **ST8511792_depth_pertrace.sgy**, which is fully compatible with the well log depth data. Script **seismic-visualisation.ipynb** is used to visualise seismic cube for both time domain and depth domain.

3. Attribute Extraction:

In this step, we generated several important seismic attributes such as **instantaneous amplitude, frequency, phase, reflection strength, RMS amplitude, sweetness, thin-bed indicator, and discontinuity**.

All these attributes were extracted using the script named **extract-all-attribute.ipynb**, and the output files were saved in SEGY format (for example, discontinuity_vectorized_fast.sgy).

4. Machine Learning Models

- **Random Forest (RF)**
- **CatBoost (CB)**
- **Ensemble Strategy**: Combines RF and CB predictions for enhanced accuracy.

5. Prototype/Implementation

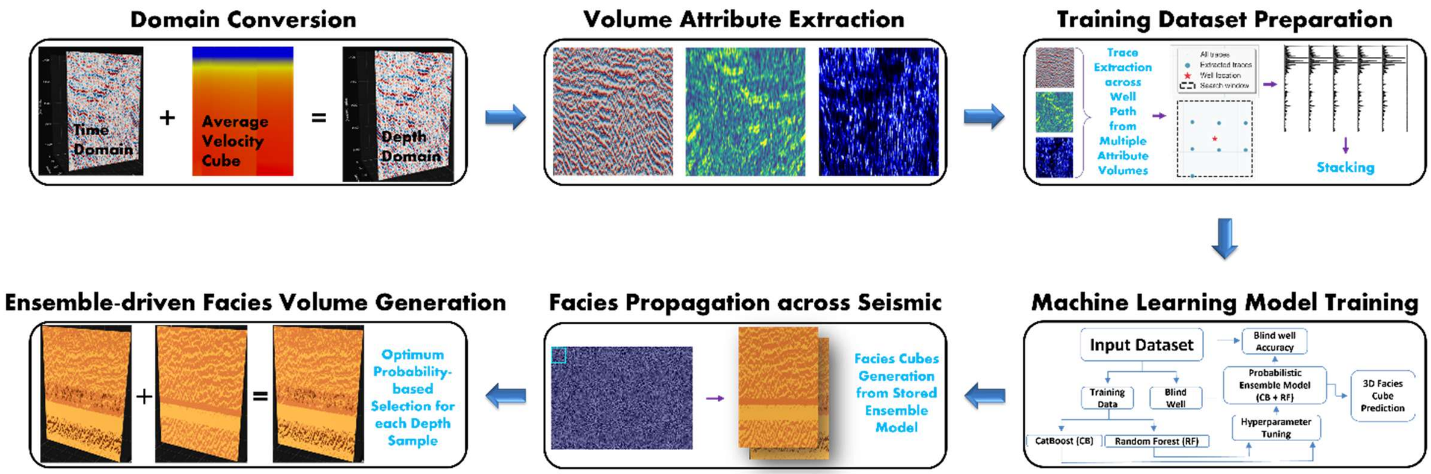

Fig 01: Workflow to generate 3D Facies cube using Multi-Attribute seismic and well data.

6. Repository Structure and Module Descriptions

Folder	Purpose	Key Components
Well-Data-Analysis/	Well log processing and visualization	•Well log files (LAS/ASCII) • well-plot.ipynb
Depth-domain-conversion/	Time-to-depth seismic conversion	• seismic-depth-domain-conversion.ipynb •Average_Velocity_cube.segy •Header modification scripts • Visualization tools
Attributes/	Seismic attribute extraction	•extract-all-attribute.ipynb 10 attribute volumes: - RMS amplitude - Instantaneous frequency/phase - Discontinuity/coherence - Amplitude gradients (X/Y) - Sweetness, thin bed, reflection strength
Attribute+Training_data_preparation/	Training dataset creation	•trace-along-well.ipynb • trace-visualisation.ipynb • 10 attribute volumes • Trace extraction along wells
Model-Training+Facies-Propagation/	ML-based facies prediction	• catboost.ipynb • random_forest.ipynb • Catboost + Random Forest.ipynb • Ensemble modelling

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FINAL-SUBMISSION [SSH25-00247]
│
├──┬ Well-Data-Analysis/
│  ├── Wells/                    # Well log files (LAS/ASCII format)
│  └── well-plot.ipynb               # Well log visualization
│
├──┬ Depth-domain-conversion/
│  ├── seismic-depth-domain-conversion.ipynb  # Main depth conversion Script
│  ├── Average_Velocity_cube.segy      # Input velocity model
│  ├── ST8511792.segy              # Original time-domain seismic
│  ├── ST8511792_depth_petrtrace.sgy   # Output depth-domain seismic
│  ├── TRACE-HEADER-INFO-time-domain.ipynb    # Time header inspection
│  ├── TRACE-header-depth-domain.ipynb        # Depth header modification
│  ├── seismic-visualisation.ipynb     # Seismic Cube visualization
│  └── seismic-depth-domain/seismic-view.time-domain   # Visualisation fig.
│
│
├──┬ Attributes/
│  ├── extract-all-attribute.ipynb     # Master attribute extraction script
│  ├── rms_amplitude_hanning.sgy       # RMS amplitude attribute
│  ├── instantaneous_frequency.sgy     # Instantaneous frequency
│  ├── instantaneous_phase.sgy         # Instantaneous phase
```

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│   ├── discontinuity_vectorized_fast.sgy  # Discontinuity/coherence
│   ├── relative_amp_change_X.sgy       # Inline amplitude gradient
│   ├── relative_amp_change_Y.sgy       # Crossline amplitude gradient
│   ├── sweetness.sgy                  # Sweetness attribute
│   ├── thin_bed_indicator.sgy       # Thin bed indicator
│   ├── reflection_strength.sgy     # Reflection Strength
│
├── Attribute+Training_data_preparation/
│   ├── FINAL-OUTPUT/
│   │   ├── Final_Multi_Attribute_Facies_Porosity.csv  # Final training data
│   │   ├── discontinuity.sgy              # discontinuity attribute
│   │   ├── instantaneous_amplitude.sgy     # inst. amplitude
│   │   ├── instantaneous_frequency.sgy     # inst. frequency
│   │   ├── instantaneous_phase.sgy         # inst. phase
│   │   ├── reflection_strength.sgy        # Reflection strength
│   │   ├── relative_amp_change_X.sgy       # Relative Amp. Change x
│   │   ├── relative_amp_change_Y.sgy       # Relative Amp. Change y
│   │   ├── rms_amplitude.sgy              # RMS amplitude
│   │   ├── sweetness.sgy                 # sweetness
│   │   ├── thin_bed_indicator.sgy         # Thin bed indicator
│   ├── Trace data/
│   │   ├── all_traces_with_coords_long.csv
│   │   ├── all_traces_with_coords_wide.csv
│   ├── ST8511792_depth_petrtrace.sgy      # Seismic Depth domain data
│   ├── trace-along-well.ipynb             # Trace extraction
│   ├── trace-visualisation.ipynb          # Trace visualization
│   └── trace_location_zoomed.png          # Well position image
│
├── Model-Training+Facies-Propagation/
│   ├── catboost.ipynb                 # CatBoost training
│   ├── random_forest.ipynb            # Random Forest training
│   ├── Catboost + Random Forest.ipynb  # Ensemble combination
│   ├── Facies-vis.ipynb               # Facies volume visualization
│   ├── catboost_info/                # CatBoost training logs
│   ├── Final_Multi_Attribute_Facies_Porosity.csv  # Training dataset
│   ├── Facies_Prediction_depth_inc_catboost.sgy   # CatBoost prediction
│   ├── Facies_RF_vectorized_Trial.sgy           # RF prediction
│   ├── Facies_Prediction_RF_CatBoost_Combined.sgy # Ensemble prediction
│   ├── Final_Catboost.png             # CatBoost performance plots
│   ├── Random_Forest_final.png        # RF performance plots
│   └── Ensemble_final.png             # Ensemble performance plot
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7. Results & Analysis

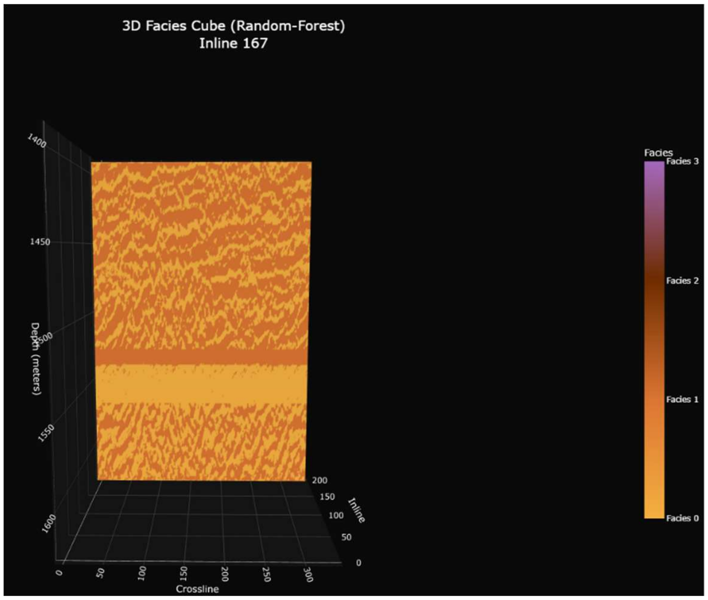

Figure 02: Random Forest predicted facies cube section(inline-167) result with blind well accuracy-0.67

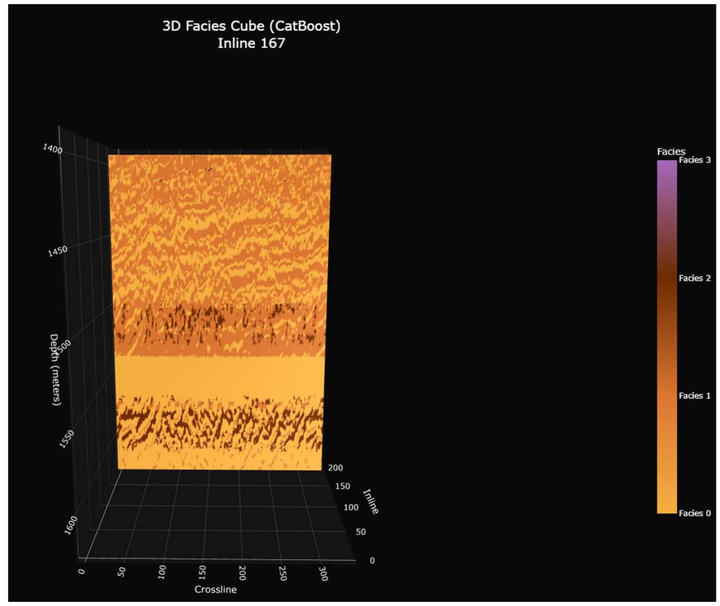

Figure 03: Catboost predicted facies cube section(inline-167) result with blind well accuracy-0.57

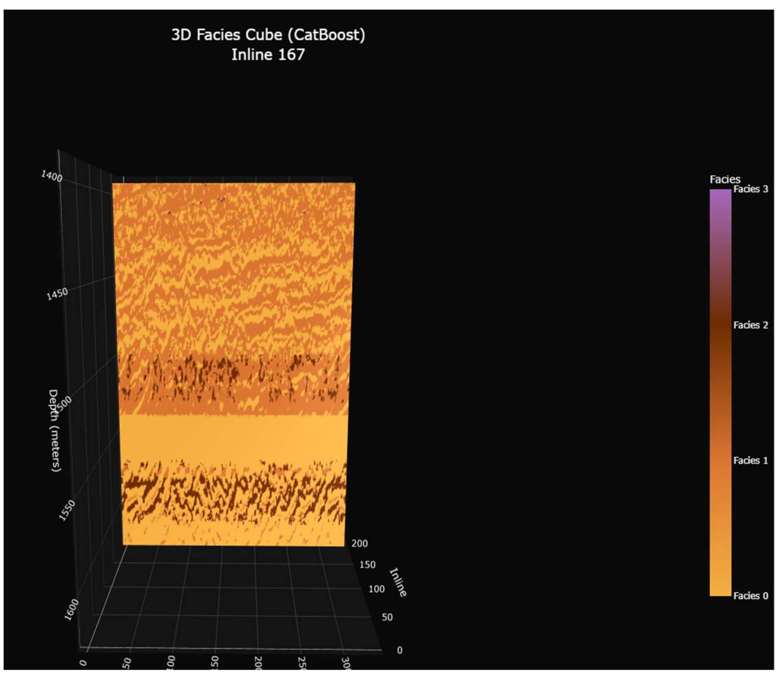

Figure 04: Catboost+Random-Forest predicted facies cube section(inline-167) result with blind well accuracy-0.59

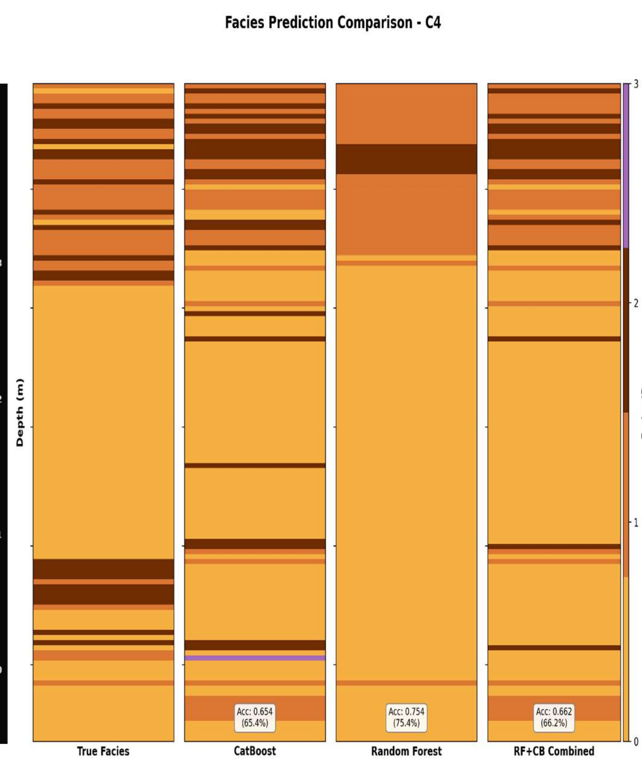

Figure 05: Comparison of facies prediction results for blind well C4 using CatBoost, Random Forest, and their ensemble model.

8. Conclusion

- Developed an integrated **ML-based workflow for automated facies prediction** using seismic and well data.
- Achieved blind well accuracy: CatBoost – 0.65, Random Forest – 0.75, Combined – 0.66 (Precision: 0.63, Recall: 0.66, F1-score: 0.65).
- CatBoost captured subtle facies (Facies–2), while Random Forest focused on dominant facies (0 & 1), while both models failed to capture the subtle variation of theFacies – 3.
- Ensemble fusion reduced uncertainty in facies propagation**.**

9. Future Scope

1. **Generate Static Reservoir Model**
    - With the derived 3D facies cube, porosity, permeability, and net-to-gross relationships can be estimated and spatially distributed.
    - We can use these properties to build a static reservoir model, providing a foundation for volumetric estimation and future dynamic reservoir simulation**.**
2. **Enhance Seismic Resolution**
    - Current mismatch between high-resolution well data (0.1524 m) and low-resolution seismic data (~4 m sampling) introduces uncertainty in facies assignment.
    - Apply **Conditional GANs (cGANs)** for seismic resolution enhancement using **synthetic seismograms** derived from **density and sonic logs** (currently missing in the dataset).
3. **Incorporate Structural Information**
    - Limited 3D seismic sections constrain structural interpretation.
    - Integrate **faults and horizon picks** to enable **structure-aware facies propagation**, preserving geological continuity and stratigraphic trends.
4. **Improve Model Diversity and Facies Balance**
    - Only **CatBoost** and **Random Forest** were used, and **facies class imbalance** (especially Facies–3) affected model learning.
    - Introduce additional **ML models** for ensemble prediction and adopt a **depth-window-based training strategy** to balance facies occurrences and enhance recognition of subtle seismic signatures.
5. **Integrate Advanced Seismic Attributes**
    - Incorporate **higher-order and physics-based seismic attributes** (e.g., spectral decomposition, curvature, coherence).
    - This will improve **facies discrimination** and enhance overall **model performance**.

10. References

- Bahorich, M., and S. Farmer, 1995, 3-D seismic discontinuity for faults and stratigraphic features: The coherence cube: 65th SEG Annual Meeting, Expanded Abstracts, 93-96.
- Gerzstenkorn, A. and K. J. Marfurt, 1999, Eigenstructure-based coherence computations as an aid to 3-D structural and stratigraphic mapping: Geophysics, 64, 1468-1479.
- Hart, B. S., 2008, Channel detection in 3-D seismic data using sweetness, AAPG Bulletin, 92, 733-742.

- Marfurt, K. J., R. L. Kirlin, S. L. Farmer, and M. S. Bahorich, 1998, 3-D seismic attributes using a semblance-based coherency algorithm: Geophysics, 63, 1150-1176.
- Oliveros, R. B., and B. J. Radovich, 1997, Image-processing display techniques applied to seismic instantaneous attributes on the Gorgon gas field, North West Shelf, Australia: 67th SEG Annual Meeting, Expanded Abstracts, 2064-2067.
- Taner, M. T., F. Koehler, and R. E. Sheriff, 1979, Complex seismic trace analysis: Geophysics, 44, 1041-1063.